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TYCOV: Test Your Cognitive Vision — An Open-Source Web Application for Human Perception of AI-Generated Images

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Software

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Summary

Advances in generative AI have made it increasingly difficult for humans to reliably distinguish between AI-generated and real images. This shift raises important questions about human perception, trust, and decision-making in digital environments.

TYCOV (Test Your Cognitive Vision) is an open-source web application designed to facilitate reproducible experiments in this space. The platform enables researchers and the public to explore how well humans identify AI-generated versus real images in a controlled, scalable, and anonymous manner.

TYCOV is not only an application but also a framework for **reproducible human–computer interaction (HCI) and AI perception studies**, supporting ethical experimentation, transparent data collection, and reproducible analysis.

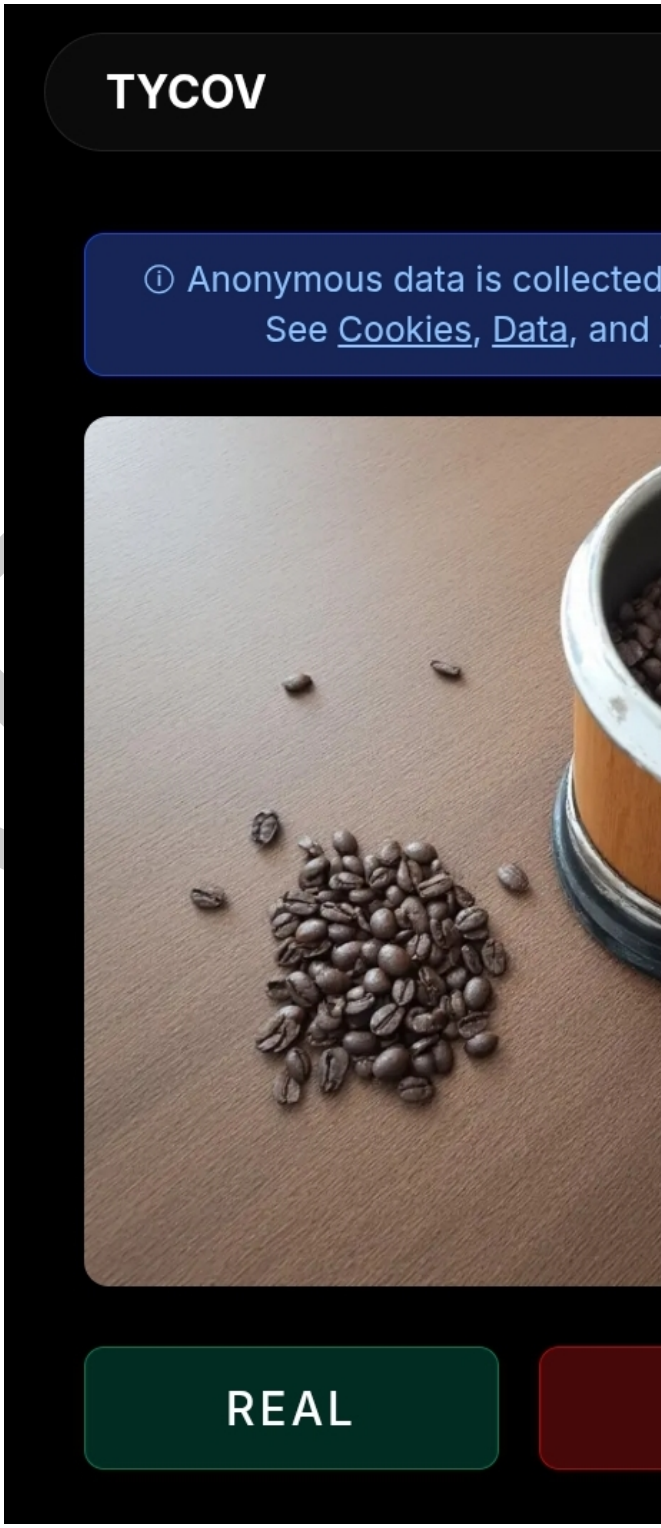
Statement of Need

While numerous studies investigate how people perceive AI-generated content, few provide **open, reusable software infrastructure** that others can adapt for their own research. Most existing works are either closed prototypes or narrowly tailored to specific datasets.

TYCOV addresses this gap by providing:

- A fully open-source **web-based platform** (Next.js + Tailwind CSS) for image-based cognitive experiments.
- Anonymized data collection with no personally identifiable information (PII).
- Integration with **Supabase** and schema management via **Drizzle ORM** for reproducibility.
- A transparent pipeline for including new datasets (AI-generated and real images).
- Built-in consent workflows for ethical experimentation.

This makes TYCOV valuable not just as a one-off experiment, but as a **generalizable tool** for HCI,



AI perception, and digital cognition research.

Implementation

The software stack includes:

- **Frontend**: Next.js (React framework) with Tailwind CSS for responsive, mobile-friendly UI.

- **Backend/Database**: Supabase (PostgreSQL) with Drizzle ORM for schema migrations and queries.
- **Deployment**: Supports both local development and cloud deployment with environment variables for Supabase integration.
- **Data Pipeline**:
  - Images organized into categories (real vs AI-generated).
  - Dataset manifest auto-generated (`manifest.json`).
  - Responses stored in the `responses` table.
  - Aggregated statistics stored in the `stats` table.

Images for the reference implementation were generated using **Stable Diffusion XL (SDXL)** (Podell et al., 2024) and **Flux.1 Schnell** (Black Forest Labs, 2024).

Reproducibility

To ensure reproducibility, the project provides:

1. **Open-source code** under the MIT License.
2. **Sample dataset** of 10 rows of data included in README.md for testing and reproducibility.
3. **Clear schema definition** for Supabase tables (`responses`, `stats`).
4. **Step-by-step documentation** for installation, dataset integration, and deployment.

This ensures that other researchers can replicate the experimental setup, swap in new datasets, and run independent studies.

Acknowledgements

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References

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